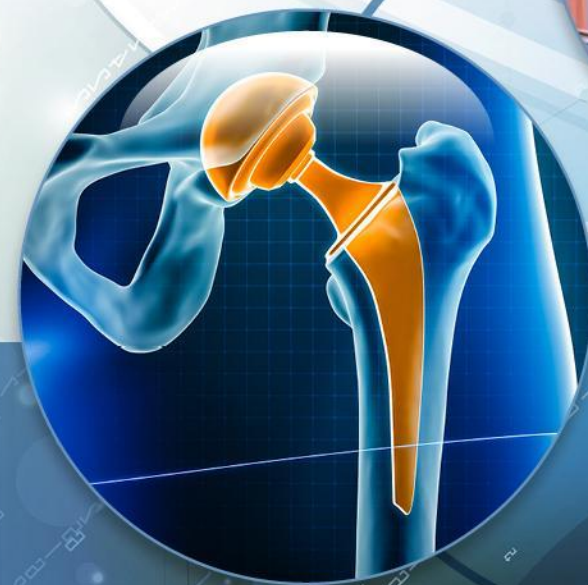


BMES

BIOMEDICAL ENGINEERING SOCIETY
2024 ANNUAL MEETING

OCTOBER 23-26, 2024
THE BALTIMORE CONVENTION CENTER
BALTIMORE, MD



BMES

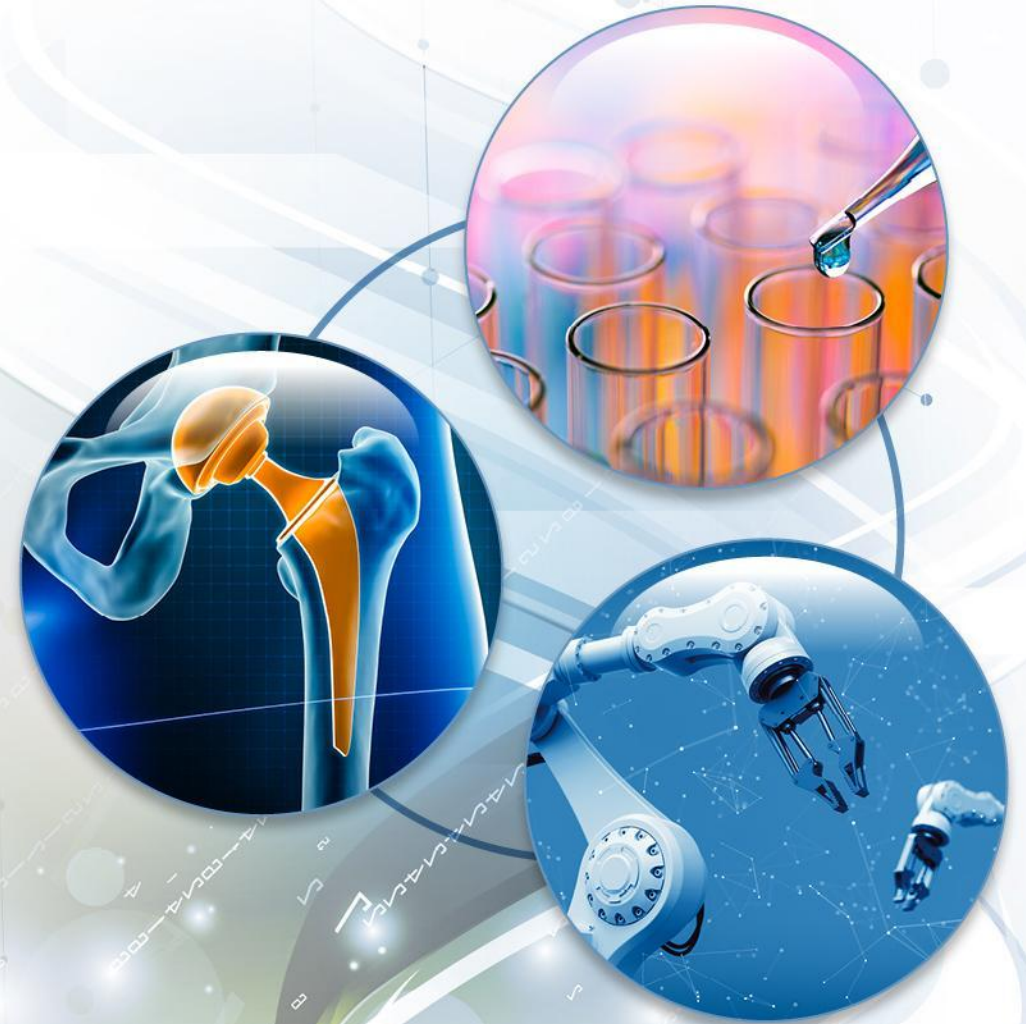
BIOMEDICAL ENGINEERING SOCIETY

2024 ANNUAL MEETING

SteadyStride: Self- Stabilizing Cane for Parkinson's Disease

Angela Altamirano¹, Reza Ashrafi¹, Andrew D'Onofrio¹, Veda Kam¹, Ally Schwartz¹, Allen Tang¹, Emma Weiss¹, Nina Williams¹

¹Cornell University College of Engineering



Device Inspiration



2011

First Sign of Symptoms

2012

Diagnosed with
Parkinson's Disease (PD)

Mo. 6

Cane User

Mo. 9

Walker User

Mo. 12

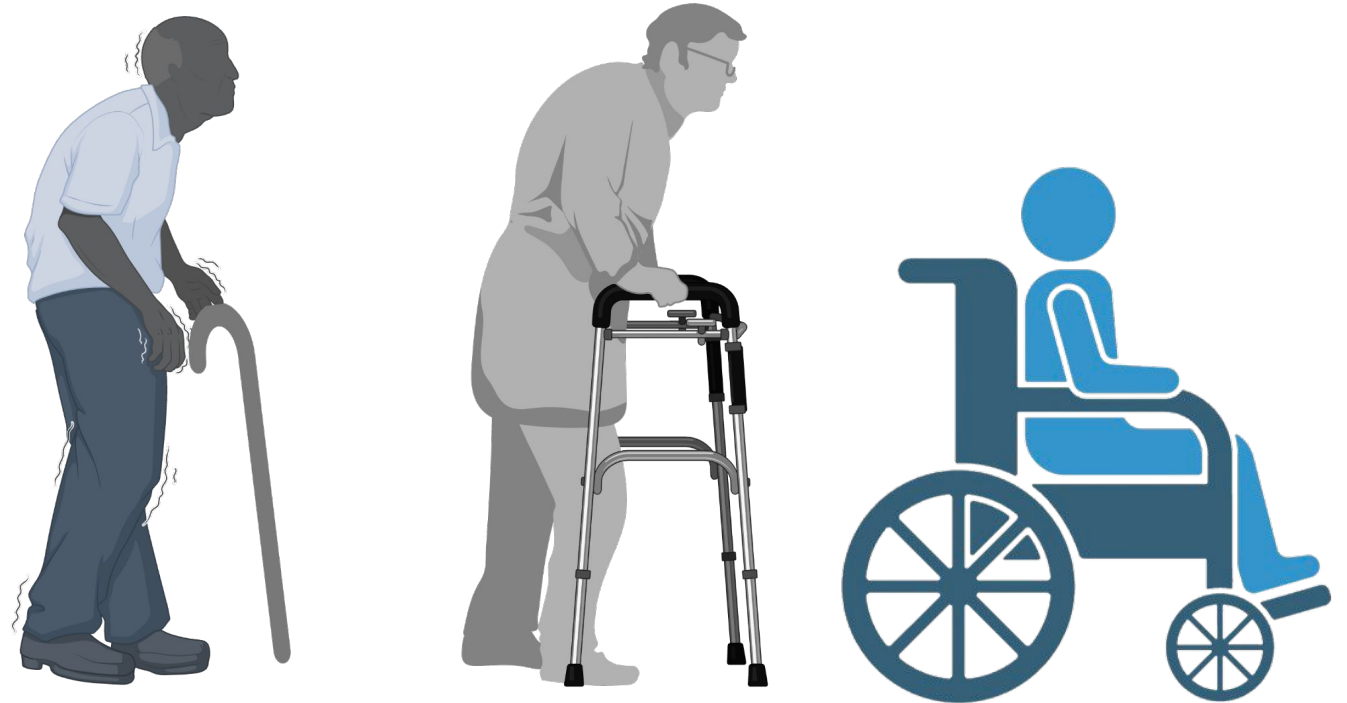
Wheelchair User

Effects of Parkinsonian Tremor



Neural Circuitry
Disruptions

Uncontrollable
Hand Tremors



Walking Assistive Device Progression

Parkinson's Patients Testimonials

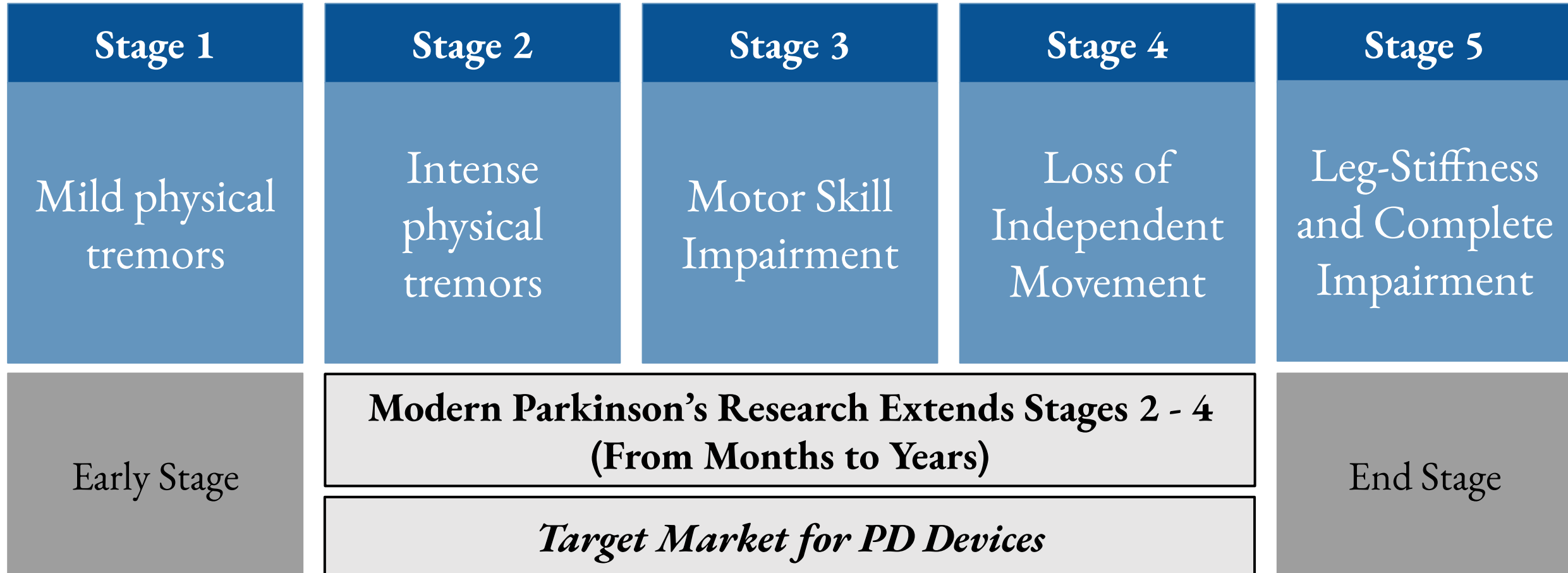
“I have thought about a walker and I get a slight **sinking feeling** ... I don't want to think about the end game.”

“**Moving is medicine** for me and it is just as important as pills. Anything that will help me to continue to walk will help me **stay a part of the world.**”

“I get scared to go to the store because I don't want to walk into a person or candy display. When I do go shopping, **my anxiety just explodes.**”

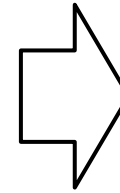


Parkinson's Disease Progression



Customer Product Needs

Device On-Market



Patient Needs

- Passive Tremor Reduction
- Discrete Profile
- Physically Supportive
- Stability while Walking
- All-terrain Mobility



STEADY *STRIDE*

A Confident Self-Stabilization Solution for
Parkinson's Patients

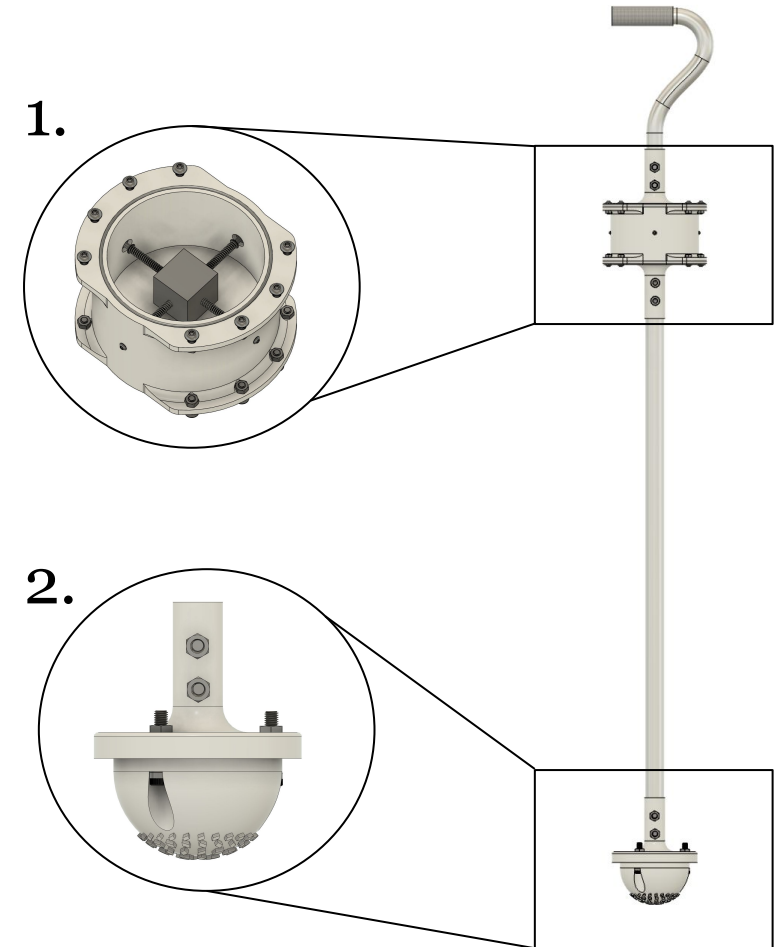
Sleek Design - Passive Damping - All-Terrain

1. Tuned Mass Damper (TMD)

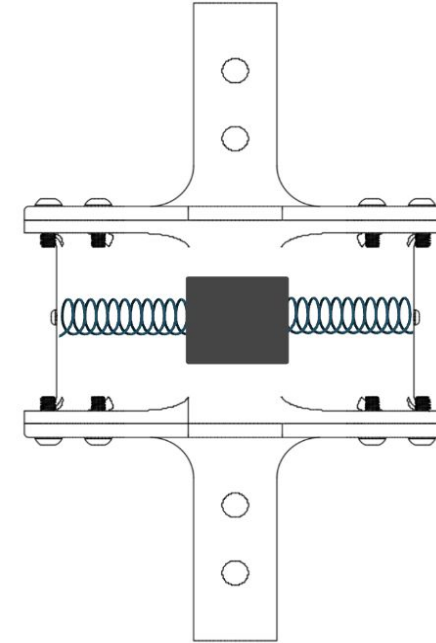
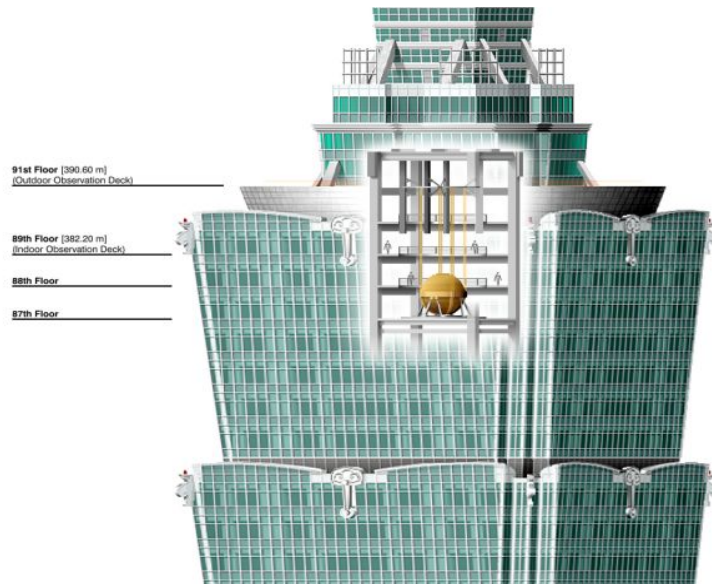
Restores Equilibrium Position

2. Dynamic Stabilizing Base

Flexible Design, Treaded for All-Terrain



Tuned Mass Damper (TMD) Theory



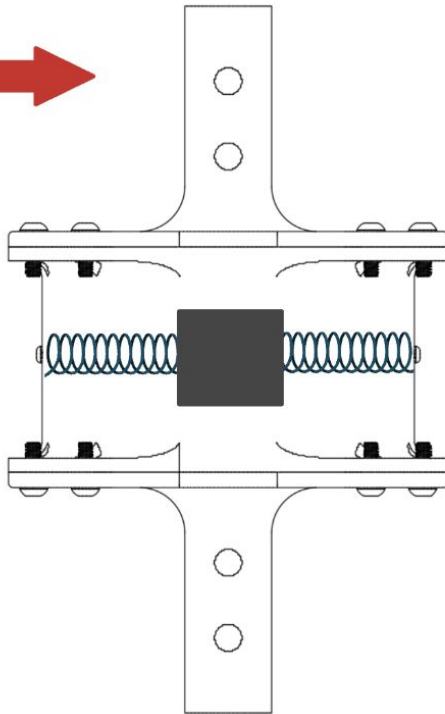
Taiwan's tallest building survived the earthquake¹: Pendulum TMD

Parkinsonsian tremor mechanically mitigated during gait: Lateral TMD

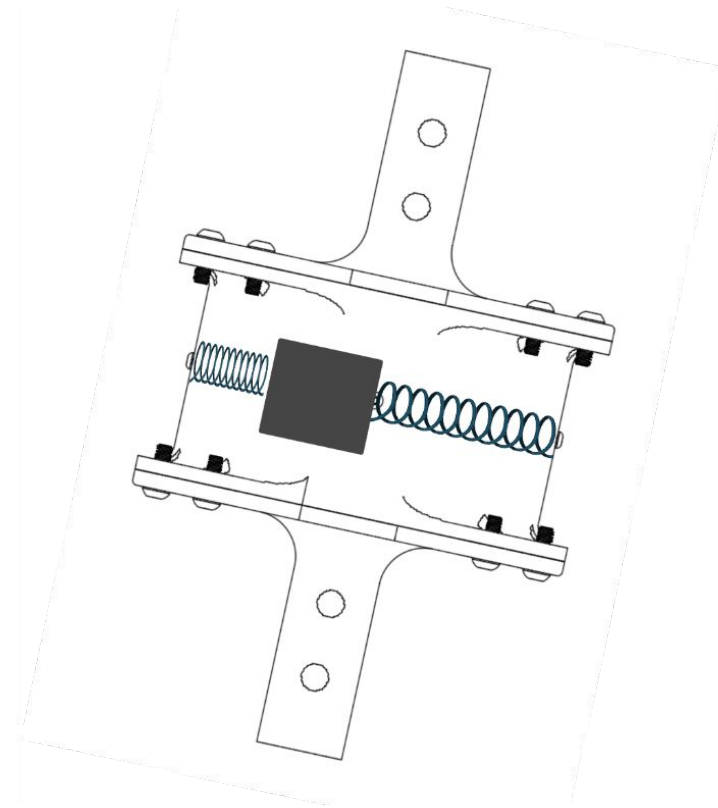
Source: The Washington Post¹

Tuned Mass Damper (TMD) Usage

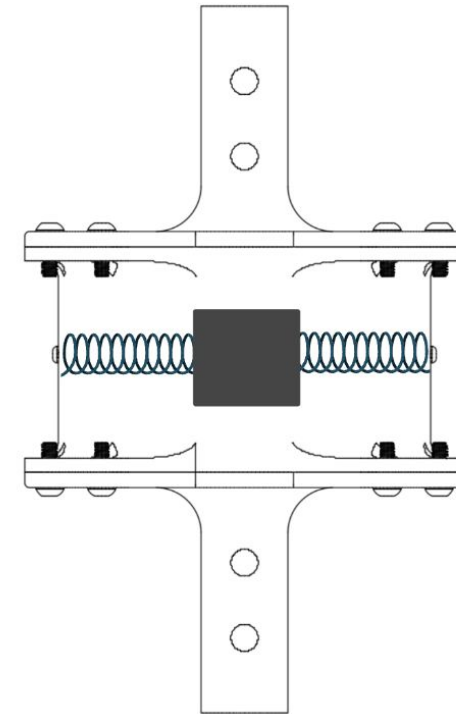
Applied
Tremor
Force



Tremor Occurs

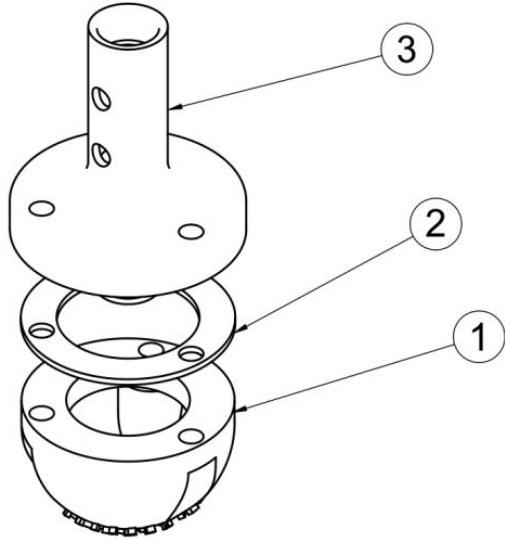


Mass Oscillates



Equilibrium Restored

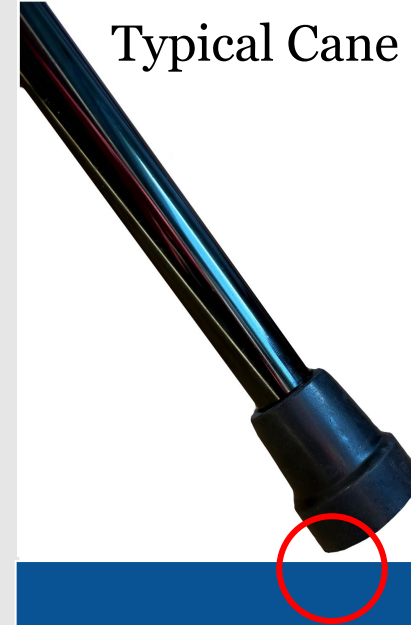
Dynamic Stabilizing Base



Non-Slip, Treaded Base for
Stability Across All-Terrain

Cane Base

Typical Cane



Unsteady

SteadyStride



Steady

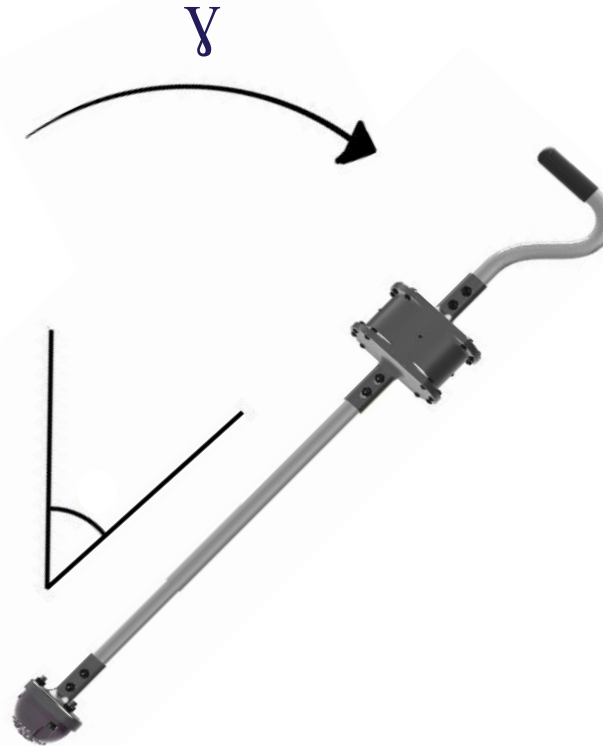
Expected Device Performance

Independent Var.

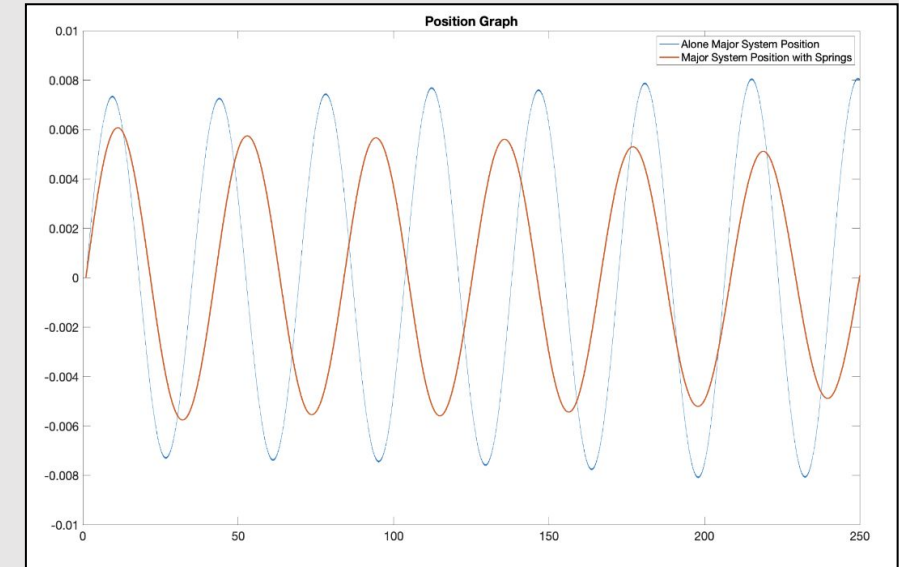
- Mass of TMD
- Spring Const. of TMD

Dependent Var.

- Angular Displacement (γ) of Total Cane



System Model:
 $\gamma = f(m, k)$



Position vs. Time of a Ideal Mass and Spring Constant Combination

IRB Study

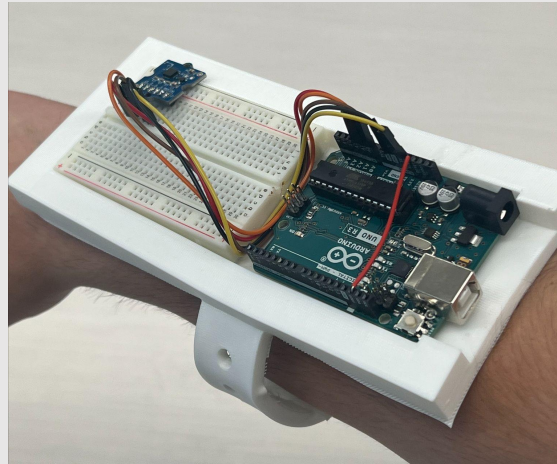


SUNY Cortland Biomechanics Lab, Dr. Bauer

*pictures provided with
participants' consent

Trials and Data Collection

Data
Collection
Device:
Accelerometer
Data



**450 Points Average Per Trial*



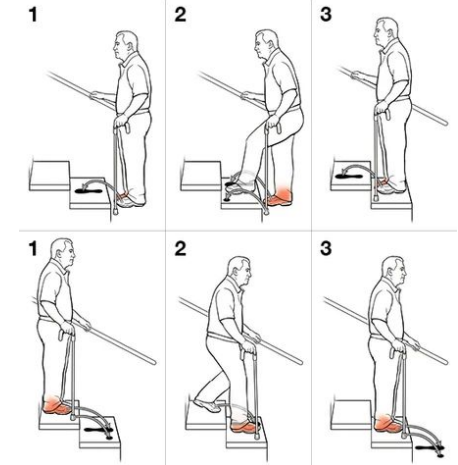
Simple Motions

B: No Motion

T1: Straight

T2: Left Turn

T3: Right Turn



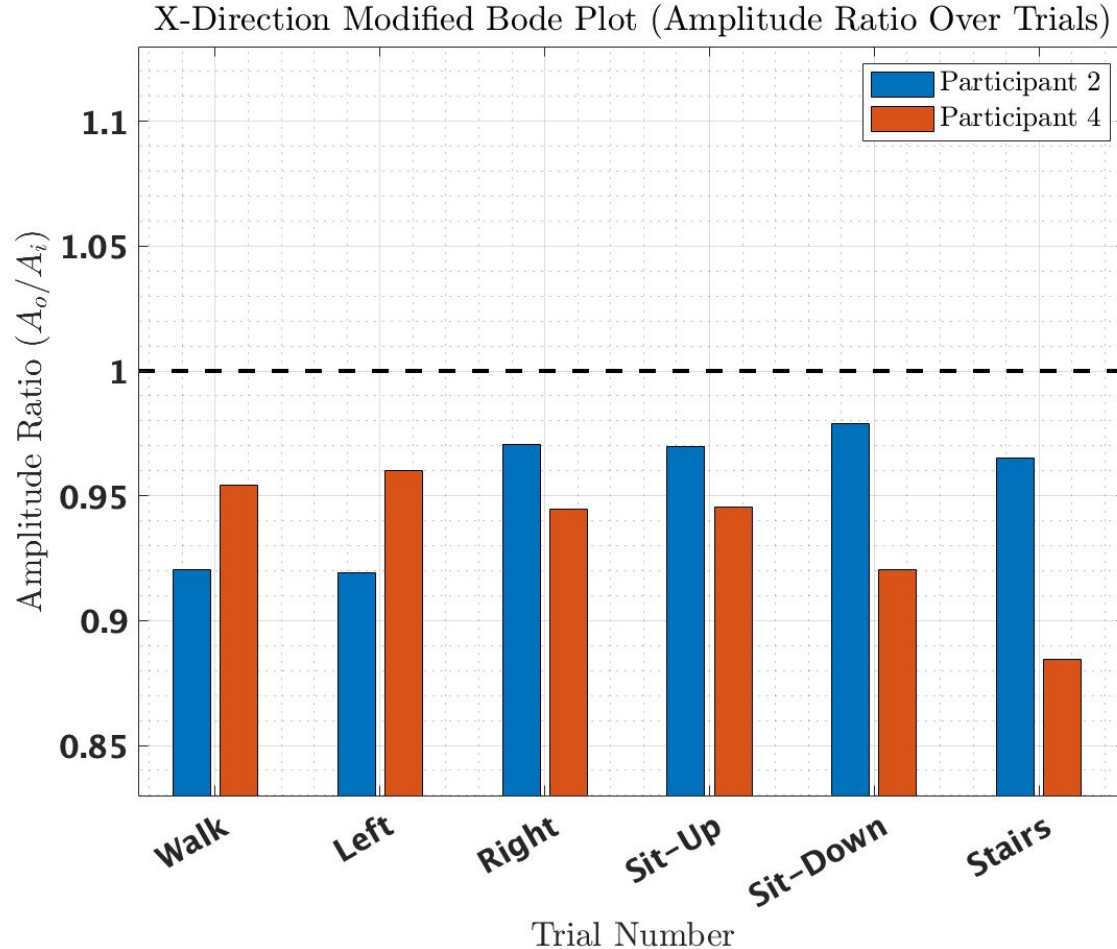
Complex Motions

T4: Sit Up

T5: Sit Down

T6: Stairs

Experimental Device Performance



$$\frac{A_o}{A_i} = \frac{\text{Tremor Amp. w/ SteadyStride Cane}}{\text{Tremor Amps w/ Base Cane}}, \text{ where}$$

$$\frac{A_o}{A_i} > 1, \text{ SteadyStride is Less Stable}$$

$$\frac{A_o}{A_i} = 1, \text{ SteadyStride has No Effect}$$

$$\frac{A_o}{A_i} < 1, \text{ SteadyStride is More Stable}$$

12.25% max decrease
of propagation in the x-direction with
the experimental cane as compared to
a standard cane for *viable* patients

Future Development

Damper Fluid: Kinetic Energy Dissipation

Model Parameters: Height & Weight

Device Specification: TMD Tuning by Weight Tier

Multi- Symptom Care: Freezing of Gait and Postural Instability





Comparison to On-Market Products

Assistive-Devices for Parkinson's Patients on Market

<i>Products</i>	<i>Passive Reduction</i>	<i>Discrete</i>	<i>Price (No Insurance)</i>	<i>Terrain</i>	<i>Target Tremors</i>
Cala kIQ	✗	✓	\$3200.00	All	Hand
Saeboglove	✓	✗	\$350.00	All	Hand
Rollz Motion Rhythm	✓	✗	\$2000.00	Some	None
SteadyStride	✓	✓	\$220.00	All	All



Prototyping & Manufacturing Budget

Prototype Manufacturing Methods

Manual Milling

Laser Cutting

3D Printing

Small Stock

Proto. Costs:

\$227.88

High-Volume Manufacturing Methods

CNC Milling

Assembly Lines

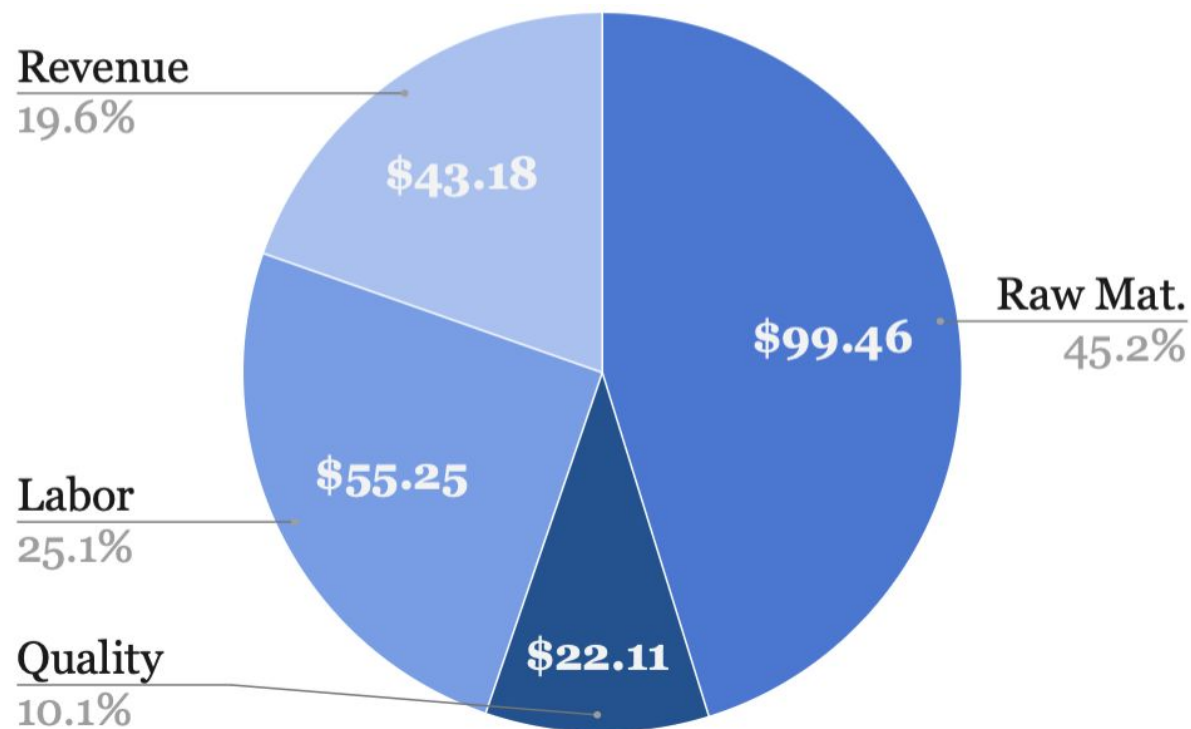
Urethane Casts

Bulk Stock

Manufact. Costs:

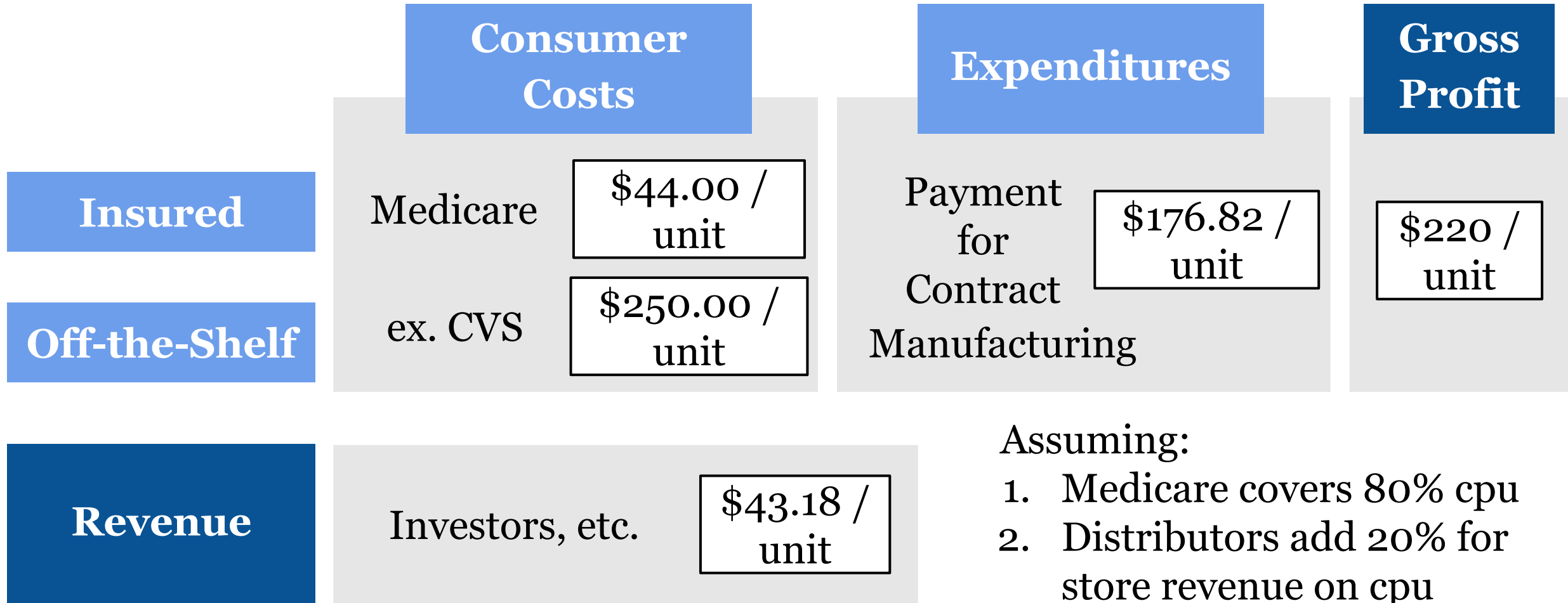
\$99.46

Product Cost Breakdown



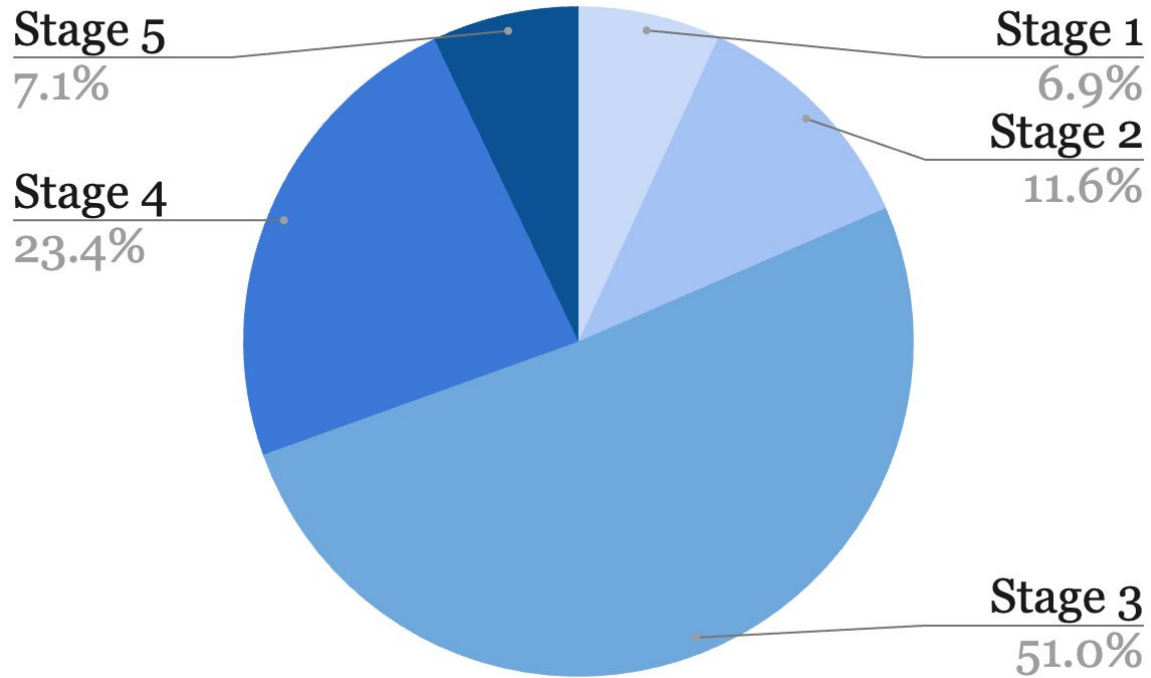
Raw Material	\$99.46
Manufacturing	\$77.36
Revenue (~20%)	\$43.18
Base Cost	\$220.00

Business Model



Anticipated Market Size

User base of 850,600 Parkinson's patients in the United States



Market Size,
Stages 2-4²

736,160

Market
Value, North
America³

\$1.31 Billion

Source: Parkinson's Foundation²
Grand View Research³

Protection Plans and Regulatory Pathway

Page 1 of 2

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www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION # 63/704,019	RECEIPT DATE / TIME 10/06/2024 10:22:01 PM Z ET	ATTORNEY DOCKET # -
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Title of Invention
SteadyStride

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT # -
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FDA Classification

Class 1 Device

Regulatory Pathway

De Novo
Classification

Prescription Model

Multiple Products
based on Weight of
the Patient

Ex.

Typ. 1: 80 - 100 kg
Typ. 2: 100 - 120 kg
Typ. 3: 120 - 140 kg

✓ SteadyStride has filed for a
provisional patent.

Milestones and Next Steps

Milestones

April 15th, 2024: IRB Trial Validation
October 6th, 2024: File Provisional Patent

Next Steps *With Investor Funding*

December, 2024: Second-Round Testing with
Updated Device

Integrate Testing With Damping Fluid

January, 2025: Start FDA Approval Process
August, 2025: Launch in 10 stores in Upstate
New York

October, 2025: Launch in Tri-State Area

Acknowledgements



Dr. Jeffrey Bauer
*Professor at SUNY
Cortland
Kinesiology
Department*



Dr. James Antaki
*Professor at Cornell
University
Meinig School of
Biomedical Engineering*

Special Thanks to Additional Members

1. Diana Michaelson - *Cornell DEBUT*
2. Michael Volintiru - *Cornell DEBUT*
3. Skyler Brown - *Cornell DEBUT*
4. Madison Cohen - *Cornell DEBUT*
5. Senegal Mabry - *Cornell University Affect and Cognition Lab*

Join Our Team!



Become an investor or technical advisor of the SteadyStride Team:

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